

Aspen Johnson

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EDUCATION

Valdosta State University

Incoming B.S. in Data Science: Computational Science & Engineering Track; Minors: Physics and Mathematics

Palm Beach State College

A.A. in STEM: Mechanical & Aerospace Engineering Track

Completed May 2026, GPA: 3.20

EXPERIENCE

Collegiate Women's Basketball Player

June 2022 – Present

Mercer University; Valdosta State University

NCAA Division I / Division II

- Commit 20–25+ hours per week to practices, strength training, film study, travel, competition, recovery, and team responsibilities while balancing full-time academic and research commitments.
- Develop leadership, discipline, time management, communication, resilience, and performance under pressure through year-round collegiate athletics.
- Represent university athletic programs in competitive, team-based environments requiring accountability, preparation, adaptability, and consistent execution.

Physics Undergraduate Researcher

Mar 2026 – Present

Valdosta State University

On-Site/Remote

- Studied the Velocity Autocorrelation Function to analyze how particle velocities evolve over time in molecular dynamics trajectories.
- Investigated how velocity data can be modeled using Gaussian distributions to better understand equilibrium behavior in simulated systems.
- Review model outputs, visualizations, and research notes for accuracy, clarity, reproducibility, and consistency before faculty discussion.

Econometrics Undergraduate Researcher

Jan 2025 – Dec 2025

Mercer University

On-Site/Remote

- Co-authored [An empirical application of improved gradient scaling for score-driven volatility filters](#), published in *Applied Economics Letters*, Taylor & Francis, 2025.
- Contributed to data curation, formal analysis, methodology, software implementation, validation, visualization, and manuscript preparation for empirical testing of the Hessian-filter scaled Beta-t-EGARCH volatility model.
- Analyzed financial time-series data across Bitcoin, S&P 500, Gold, U.S. REITs, and CHF/JPY from 2010–2025 using R, GAUSS, Excel, and Bloomberg Terminal data.
- Selected as 1 of 15 research papers to present a lecture at the 2026 Atlanta Research Conference.

Financial Analyst Intern

May 2025 – Dec 2025

[Zinzino USA](#)

Remote

- Reduced chargeback processing time by 87% through workflow digitization, documentation streamlining, and process-improvement analysis.
- Created reference documentation to support onboarding, data quality, process consistency, and cross-functional accuracy.
- Reviewed finance-related documentation and case details for completeness, accuracy, and clarity before sharing guidance with internal stakeholders.

PROJECTS

Econometric Modeling in Complex Physical Systems | *Python, R, Time-Series Analysis, Volatility Modeling*

- Investigating whether GARCH-family and long-memory time-series models can be adapted to complex physical systems exhibiting clustering, persistence, uncertainty, or volatility-like behavior.
- Sourcing and structuring geophysical datasets from USGS, NOAA, and NCAR, including seismic activity, geomagnetic storms, and solar wind turbulence, for empirical model validation.
- Exploring how financial-market modeling tools may support broader analysis of event clustering, infrastructure-adjacent risk, system behavior, and reliability-focused interpretation.
- Documenting model assumptions, limitations, data quality issues, and interpretation risks to support reproducible research workflows and avoid overclaiming predictive results.

Assistive Hearing Technology Concept | *Accessibility Research, Signal Processing, Embedded Systems*

- Investigating a low-cost assistive hearing technology concept combining accessibility research, audio/signal processing concepts, embedded systems, and human-centered design.
- Researching how microphones, sensors, microcontrollers, and software-based filtering could support improved sound awareness in athletic events and high-noise environments.
- Documenting user needs, technical requirements, component options, and design constraints for future prototype development and applied technology testing.

DIY Cyberdeck / Portable Computing Hardware System | *Fusion 360, KiCad, PCB Design, Embedded Systems*

- Designing a modular portable computing device inspired by the [ClockworkPi uConsole](#), emphasizing enclosure constraints, component layout, usability, repairability, and field-ready portability.
- Using Fusion 360 to prototype enclosure concepts, internal spacing, mounting points, and component placement for a compact hardware system.
- Learning KiCad workflows for schematic capture, PCB layout, component selection, and hardware design documentation.
- Applying electronics, electromagnetism, embedded systems, and hardware/software integration concepts to strengthen applied engineering and technical documentation skills.

LEADERSHIP, COMMUNITY ENGAGEMENT & EXTRACURRICULARS

Weekly Physics Study Group Member | *Studying Helliwell's Modern Classical Mechanics*

Read2Succeed Tutor | *Supported elementary literacy development and student confidence at Matilda Hartley Elementary*

AWS Campus Prep Series Participant | *Completed career and cloud-focused webinar series*

TECHNICAL SKILLS

Programming & Analysis: Python, R, GAUSS, Java, C/C++, JavaScript, Octave, Excel

Data Science & Modeling: Data cleaning, exploratory data analysis, time-series analysis, statistical modeling, model validation, data visualization, forecasting concepts

AI/Data Review: AI response evaluation, prompt testing, data annotation, data labeling, data quality review, pattern recognition

Engineering & Prototyping: Fusion 360, KiCad, PCB design fundamentals, circuit design fundamentals, embedded systems concepts, technical diagrams

Tools: NumPy, Matplotlib, Seaborn, Google Colab, $\LaTeX$, Git, CLI, VS Code, Bloomberg Terminal

Research & Documentation: Literature review, technical writing, reproducible notes, model comparison, workflow documentation